

Project design Phase

Proposed Solution

Date	28-06-2026
Project Name	Automated Network Request Management
Team ID	
Maximum Marks	

Here is the proposed architectural solution for the Automated Network Request Management (ANRM) project, detailing the core platform components, technical parameters, and their specific execution mechanics.

Proposed Architectural Solution

S.No.	Parameter	Description
1	User Intake Layer (Service Portal)	A clean, dedicated self-service catalog item in ServiceNow using dynamic UI policies. It ensures all input parameters (Source/Destination IPs, Ports, Protocols) are collected upfront with strict client-side regex checking to prevent invalid data from entry.
2	Orchestration & Logic Engine (Flow Designer)	The core execution workflow that runs immediately upon form submission. It parses the inbound variables, runs them against an automated risk evaluation matrix, and handles multi-tier conditional routing logic based on the risk tier of the request.

3	Integration Pipeline (Integration Hub)	Uses structured integration spokes to programmatically translate ServiceNow request variables into precise JSON/XML payloads. It manages safe, credentialed outbound REST API or SSH communication paths to target network controllers.
4	Target Infrastructure Fulfillment Layer	The recipient network architecture layer (e.g., Palo Alto Panorama, Cisco DNA Center, or Infoblox IPAM) that accepts the automated API commands from ServiceNow and executes the live firewall rule or routing changes without manual engineer intervention.
5	Closed-Loop Verification Engine	An automated script that executes immediately post-deployment. It triggers network verification checks to confirm active connectivity over the requested path, ensuring the implementation was successful before proceeding.
6	Configuration Track-Back (CMDB Update)	A database write-back transaction where successful network modifications trigger an automatic update to target configuration items (CIs) and their relationships, keeping the infrastructure inventory completely accurate.
7	Compliance Tracking & Audit Trail	A centralized, immutable logging process that links every single automated action—including initial user parameters, manager approvals, payload details, and raw API responses—directly onto the ticket record for foolproof compliance.

8	Lifecycle Communication Channel	Automated system event triggers that send real-time notifications (via email or corporate chat webhooks like MS Teams/Slack) to the requestor at critical milestones: submission, approval, and completion.
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